



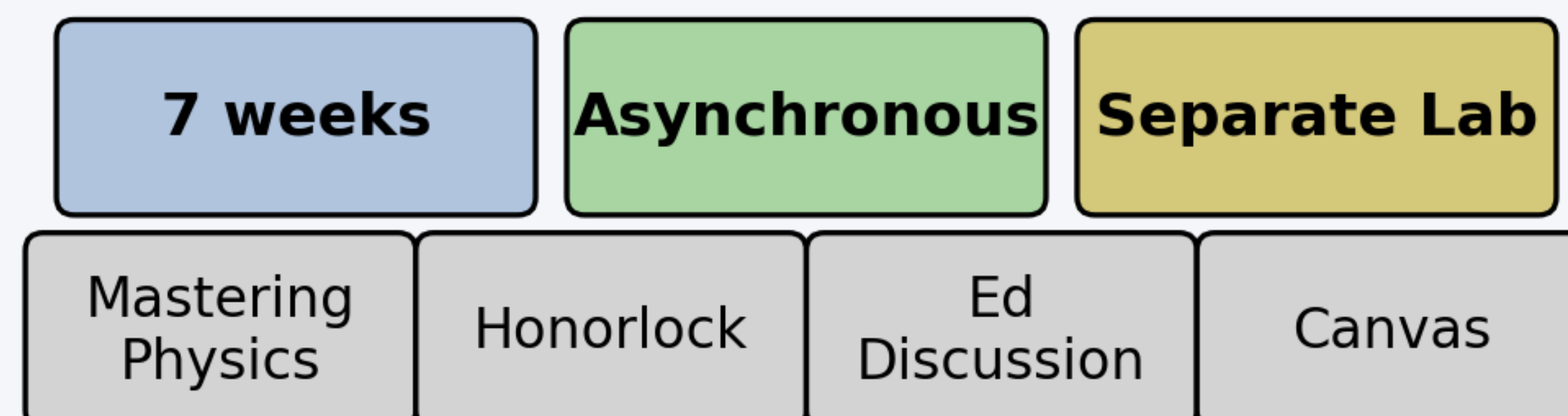
Background

Adult learners (age 25 and older) must often balance coursework with employment, caregiving, and other responsibilities. Asynchronous online courses can give them access to higher education, but this format can add many hurdles to learning.

Research Questions

- RQ1** Barriers and enablers to success in online async physics.
RQ2 Perceptions of academic integrity: proctoring (Honorlock), AI, cheating.
RQ3 How do online students compare to in-person on learning outcomes (passing rate, content learning)

Methods & Setting



- Large public university; 8-week, fully async intro mechanics; Honorlock-proctored exams = 80% of grade.
- 11 semi-structured Zoom interviews (45–60 min) in Summer 2025
- Codebook thematic analysis

Participants

All but one adult learner had full-time employment whereas traditional age students were part-time employed. The majority of adult learner also had part-time caregiving duties.

Participant Pseudonyms	Age	Gender	Race
Adult learners			
Cameron	36	M	White
Elijah	27	M	Black
Treven	27	M	White
Timothy	39	M	White
Kevin	39	M	White
Austin	46	M	White
Caleb	24*	M	White
Maria	32	W	Indigenous
Traditional age			
Sara	19	W	Asian
Samir	20	M	Asian
Maya	20	W	Asian

Barriers, Enablers and AI

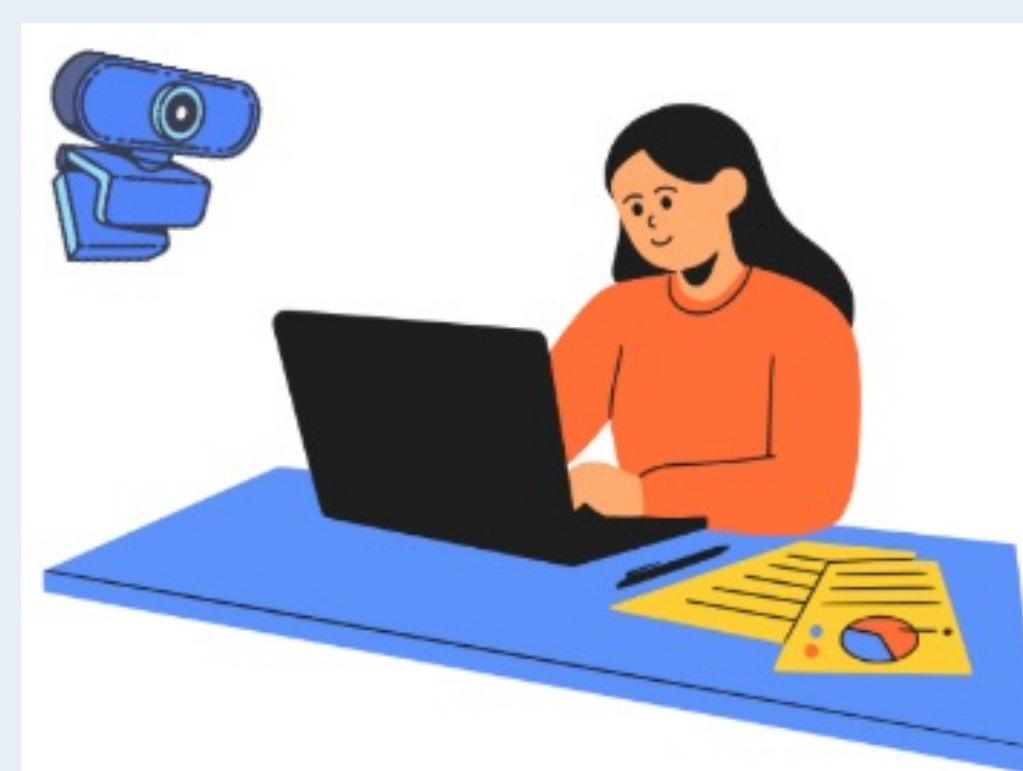
It's a balancing act: managing work, life, and school.

Self-regulation, motivation and family support are crucial.

"I feel like my life feels very busy and full and wonderful, and having the, like, asynchronous flexibility of an online program is a necessity... So I'm more than happy to trade off the joys of in person learning for, like, the efficiency and flexibility of online learning. — Kevin"



AI seen as an enabler to studying while Honorlock proctoring seen differently between adult learners and traditional age students



Adult Learners acknowledged proctoring has gaps, yet framed cheating as ultimately self-defeating.

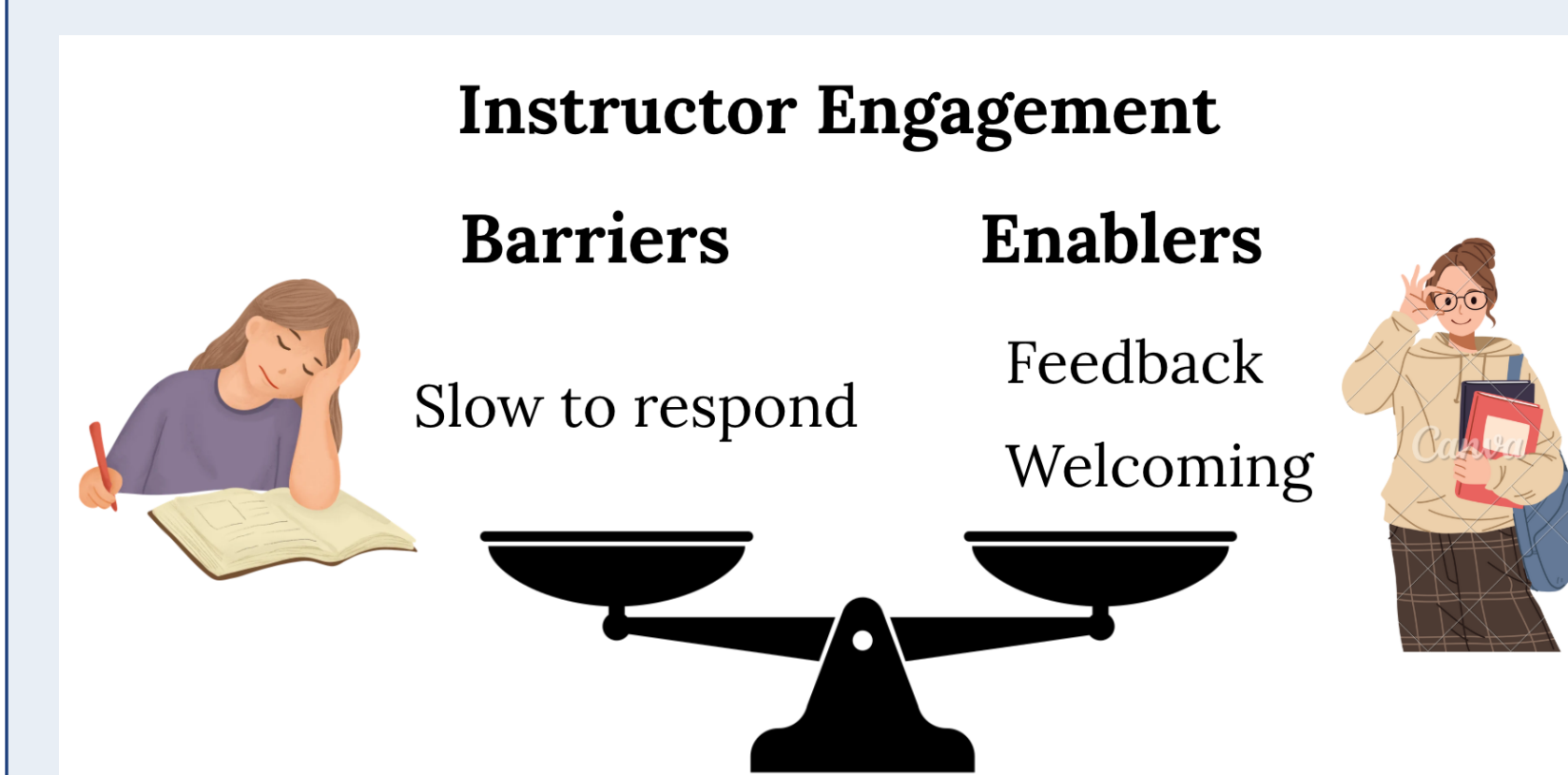
Traditional-age students talked about "messaging" (or not) with the proctoring system

"it tracks, like, your eye movement and stuff, like, personally, like, I'm, like, a little afraid, so I don't mess with it. — Maya"

"Honorlock is not a very secure system, to be completely honest. I mean, it's very easy to just hide, let's say, just hide an iPad under the table while you're doing the room scan, and then pull it out late — Samir"

Instructor's implementation and engagement lead to different experiences.

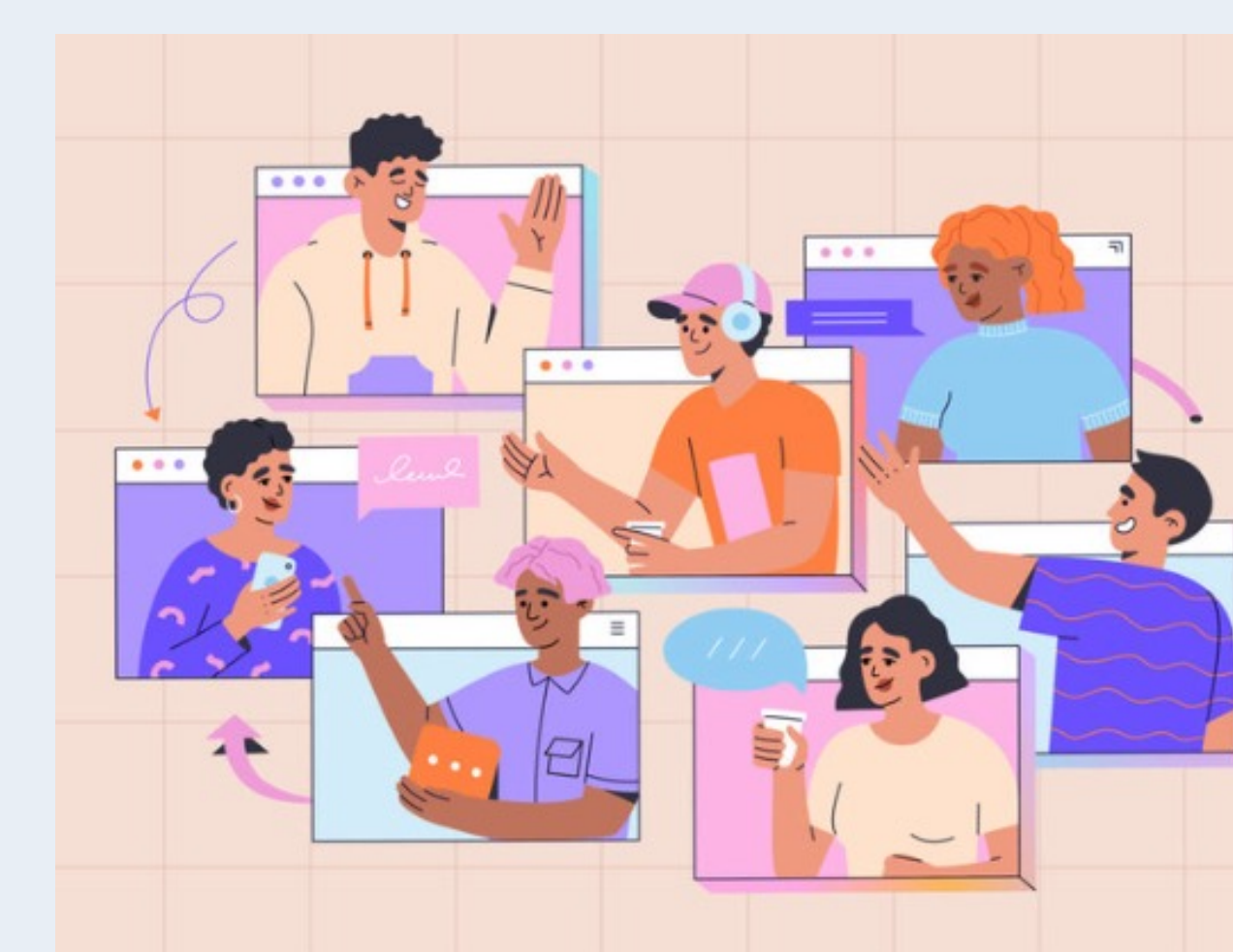
Different instructor's implementation of same course and syllabi -> noticeable difference in experiences.



"It was probably just seeing how helpful he was to all the students in those chats ... — Caleb"

Community of peers serves learning, emotional, and belonging functions for adult learners.

Connections outside of the classroom were equally if not more important than in-course interactions

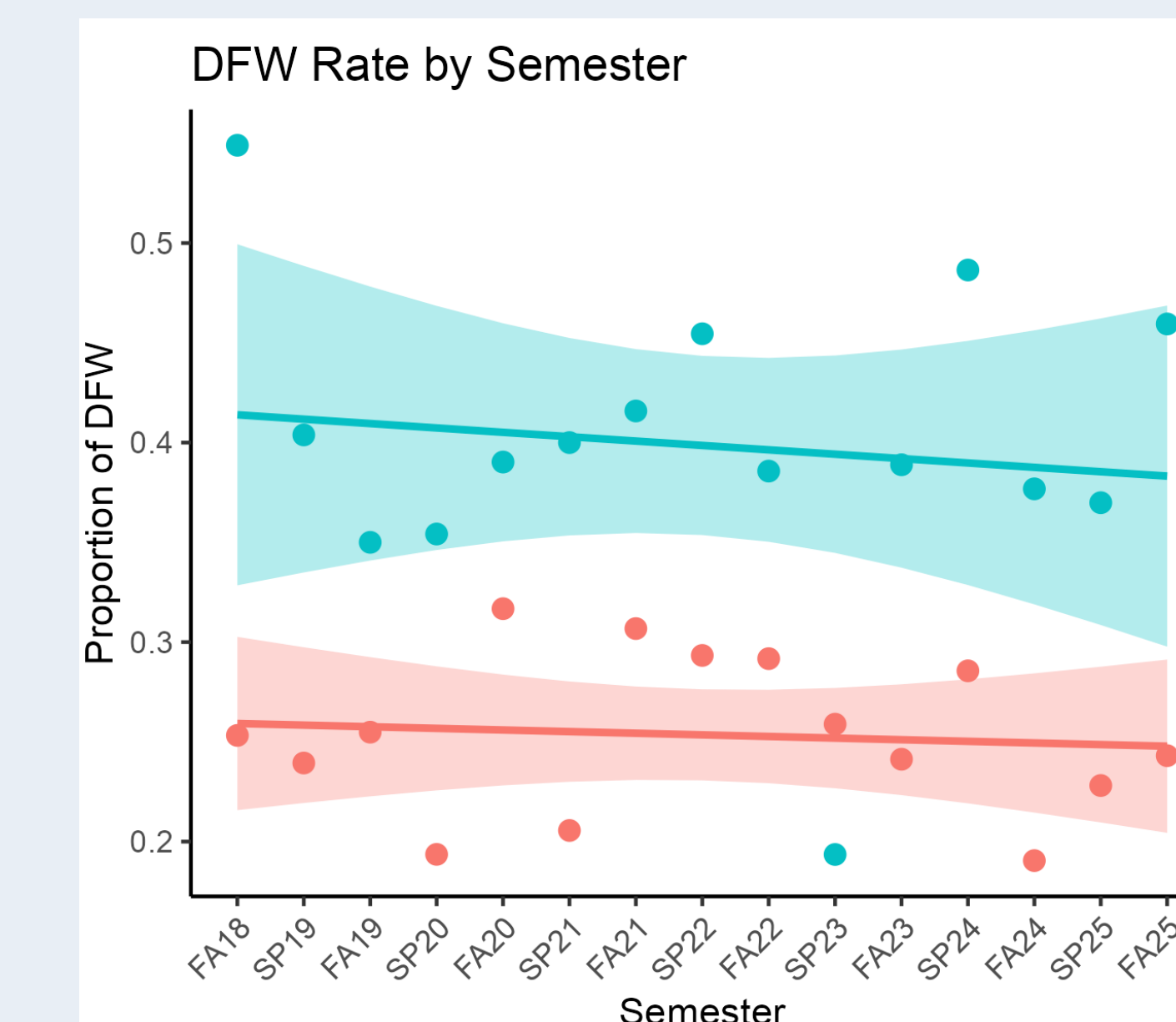


"we had a thread called Crying Corner, where people would just go in and post like, well, this is how my physics that week went. — Timothy"

Asynchronous Online vs In-person Large Enrollment

A story in three graphs from historical data at Penn State

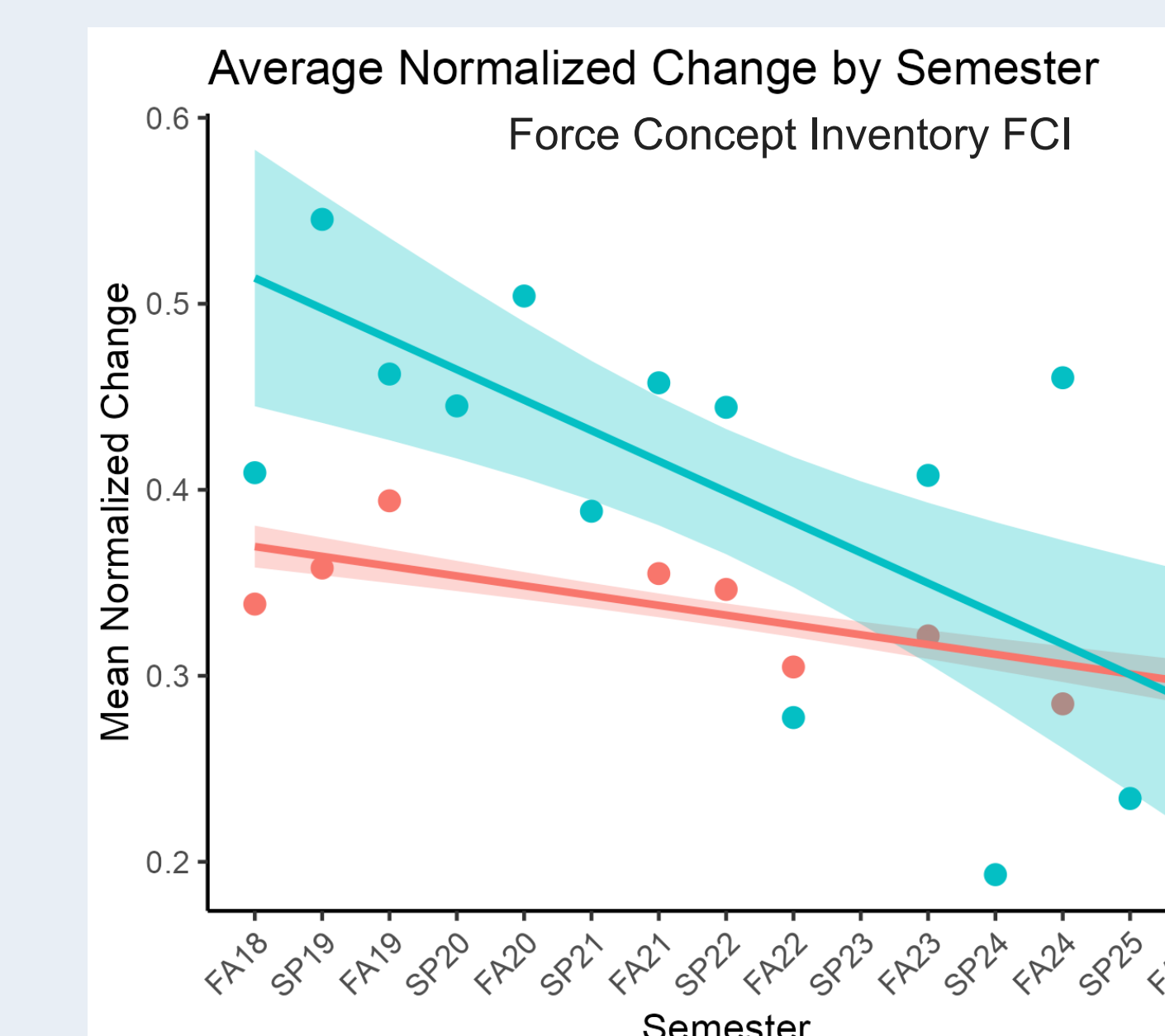
Turquoise = asynchronous online
Red = in-person – large enrollment



Compared to in-person courses, async online courses have

Higher did-not-pass rates (DFW)

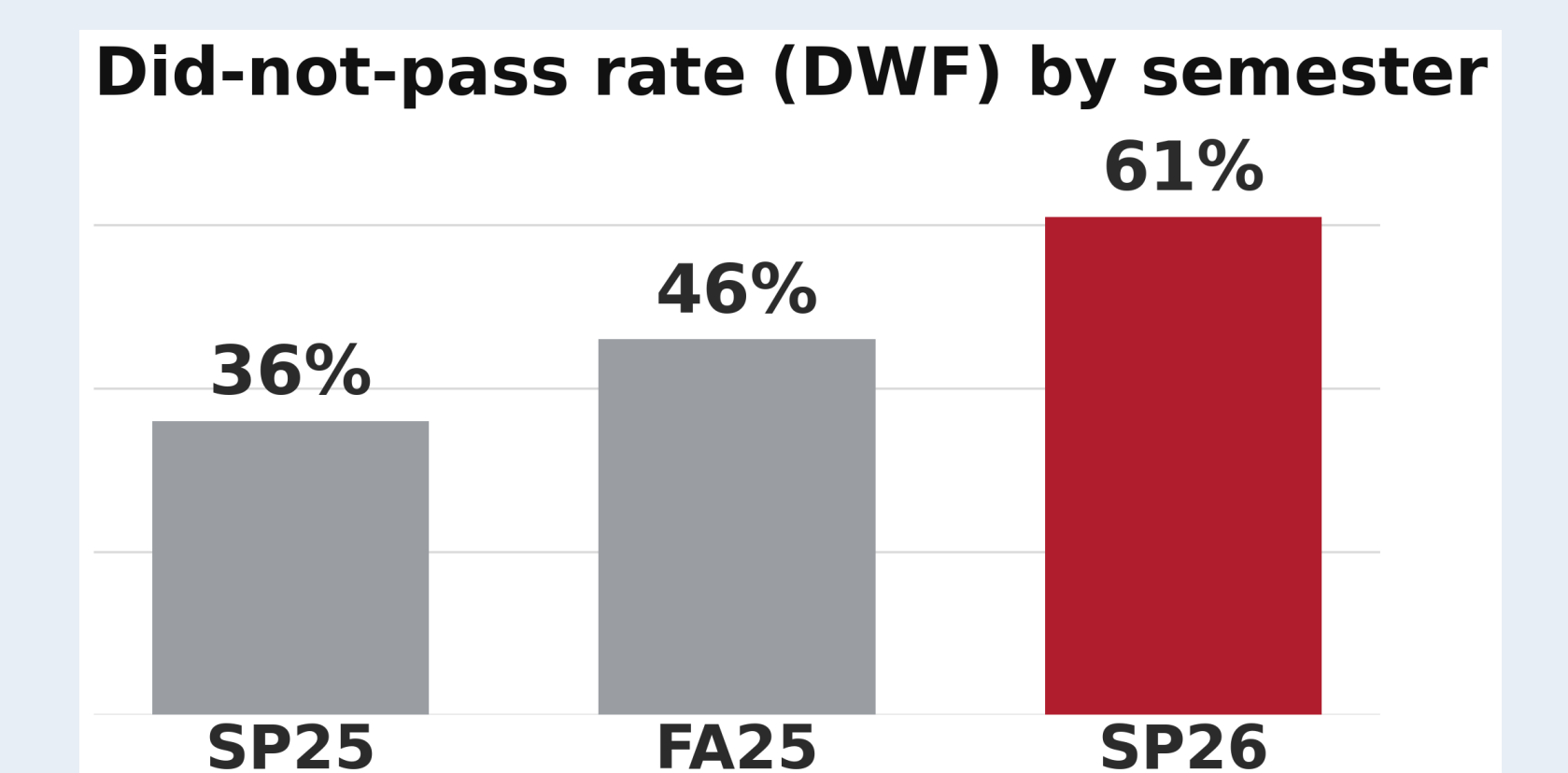
Higher* but decreasing learning gains on FCI



*comparison difficult due to attrition and population difference

Recent dramatic changes in response to perceived AI cheating

stricter proctoring rules + new oral exams led to large DFW rate increase!



Implications

- Adult learners need asynchronous learning
- Honorlock type proctoring may no longer be sufficient and threaten the integrity of the course
- Build support for self-regulated learning & load management.
- Instructor engagement matters
- Build community in and out of class